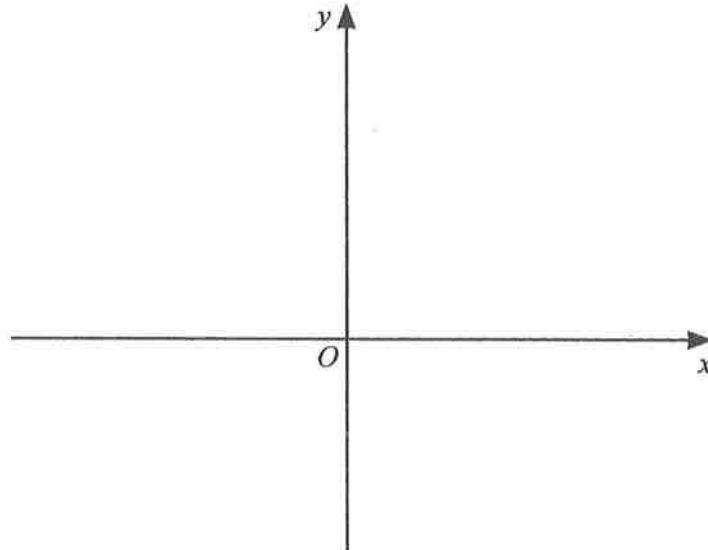


Paper 1: Basic

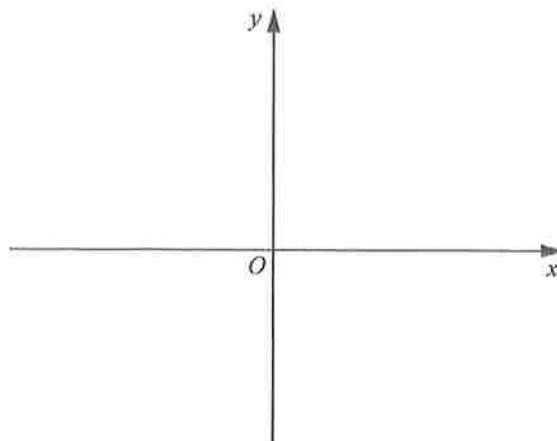
1. (a) Express $x^2 + 9x - 4$ in the form $(x + q)^2 + p$. [2]
(b) Write down the coordinates of the minimum point of the graph of $y = x^2 + 9x - 4$. [1]
[N20/I/10]

2. Sketch the graph of $y = -(x - 2)^2 + 9$ on the axes below.
Indicate clearly the coordinates of the points where the graph crosses the axes and the maximum point on the curve.



[3]
[N19/I/15]

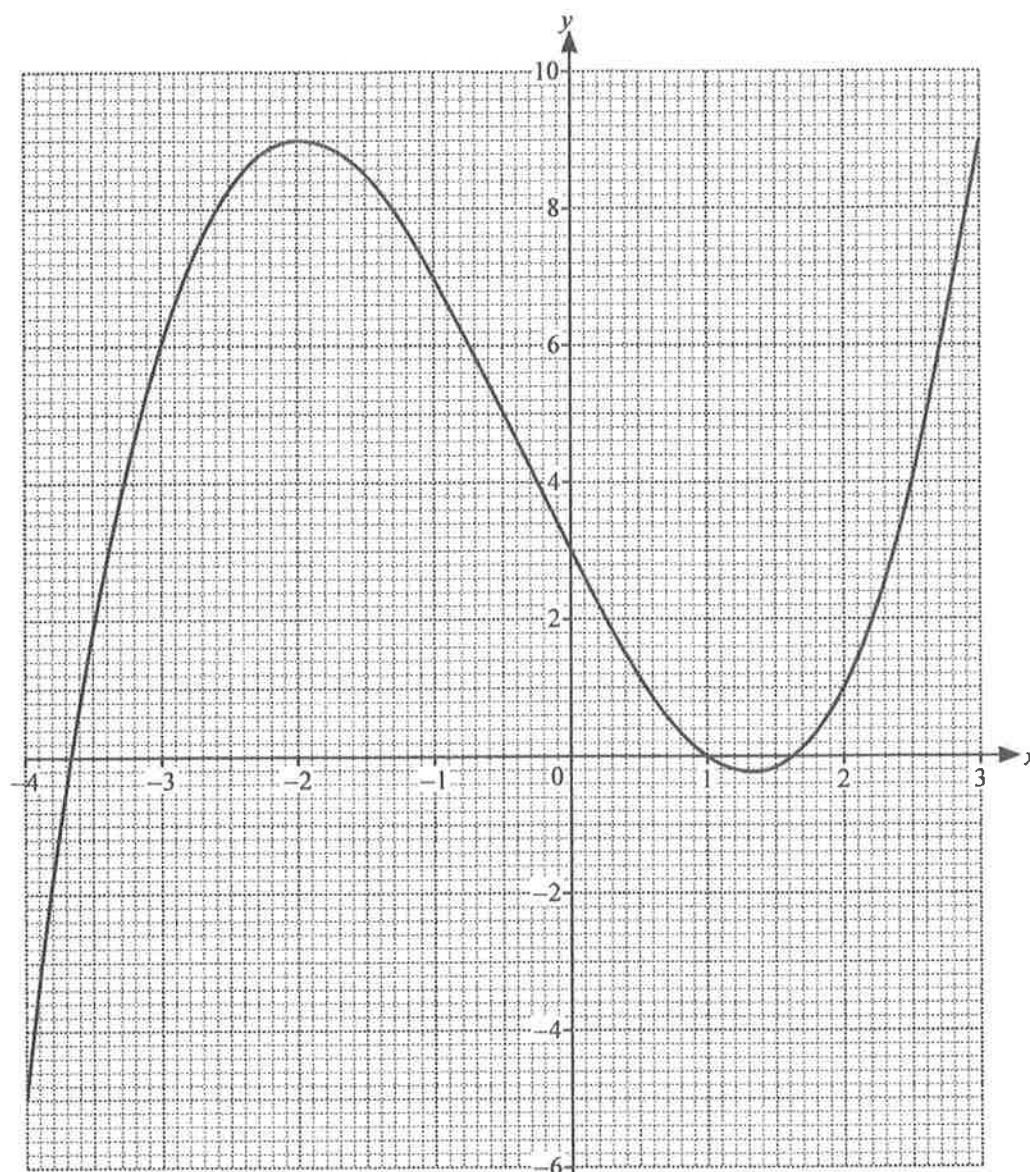
3. Sketch the graph of $y = -(x - 8)(x + 3)$ on the axes below.
Indicate clearly the values where the graph crosses the x - and y - axes.



[2]
[N17/I/4]

4. (a) Express $9 - 8x + x^2$ in the form $p + (x + q)^2$. [2]
(b) Write down the coordinates of the minimum point of the graph of $y = 9 - 8x + x^2$. [1]
[N16/I/11]

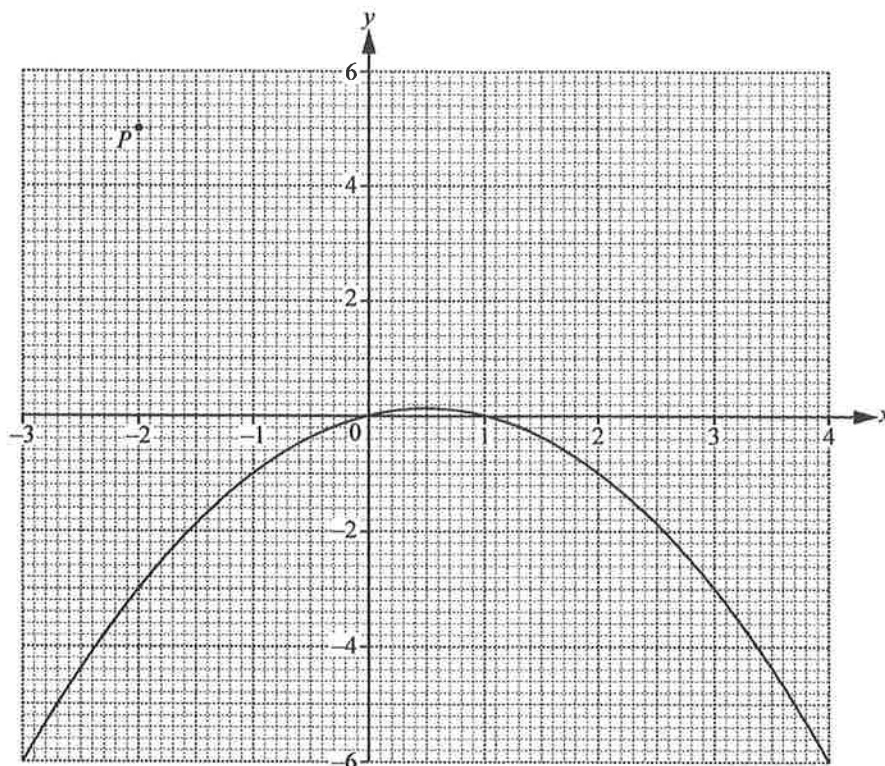
5. The graph of $y = \frac{1}{2}(x^3 + x^2) - 4x + 3$ is drawn on the grid.



- (a) By drawing a tangent, find the gradient of the curve at (2, 1). [2]
- (b) The equation $x^3 + x^2 - 6x = 0$ can be solved by drawing a suitable straight line on the grid. [2]
- (i) Find the equation of the straight line. [2]
- (ii) By drawing this straight line, solve the equation $x^3 + x^2 - 6x = 0$. [2]

[N21/I/23]

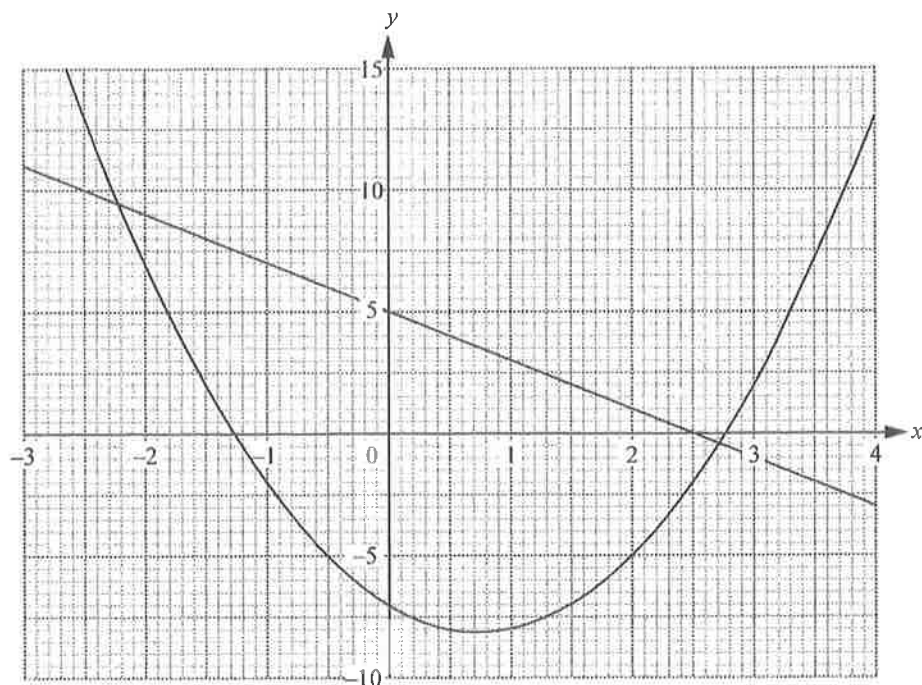
6. The graph of $y = \frac{1}{2}(x - x^2)$ is drawn on the grid.



- (a) Write down the equation of the line of symmetry of the curve. [1]
- (b) Use the graph to solve the equation $x - x^2 = -4$. [2]
- (c) The point P has coordinates $(-2, 5)$.
A tangent to the curve can be drawn so that the tangent passes through P .
- (i) Draw this tangent on the grid above. [1]
- (ii) Find the equation of this tangent. [2]

[N18/I/23]

7. The graphs of $y = 2x^2 - 3x - 7$ and $y = 5 - 2x$ are drawn on the grid.



- (a) Explain why the equation $2x^2 - 3x - 7 = k$ does not have solutions for some values of k . [1]
- (b) The points of intersection of the curve and the straight line give the solutions of a quadratic equation. Find the quadratic equation, giving your answer in the form $ax^2 + bx + c = 0$. [1]
- (c) The equation $2x^2 - 6x - 3 = 0$ can be solved by drawing a suitable straight line on the grid.
- (i) Find the equation of the straight line. [1]
- (ii) By drawing this straight line, solve the equation $2x^2 - 6x - 3 = 0$. [3]

[N17/I/21]

Paper 2: Basic

1. (i) Express $9 - 7x + x^2$ in the form $p + (q + x)^2$. [2]
- (ii) Write down the coordinates of the minimum point of the graph of $9 - 7x + x^2$. [1]

[N19/II/1(c)]

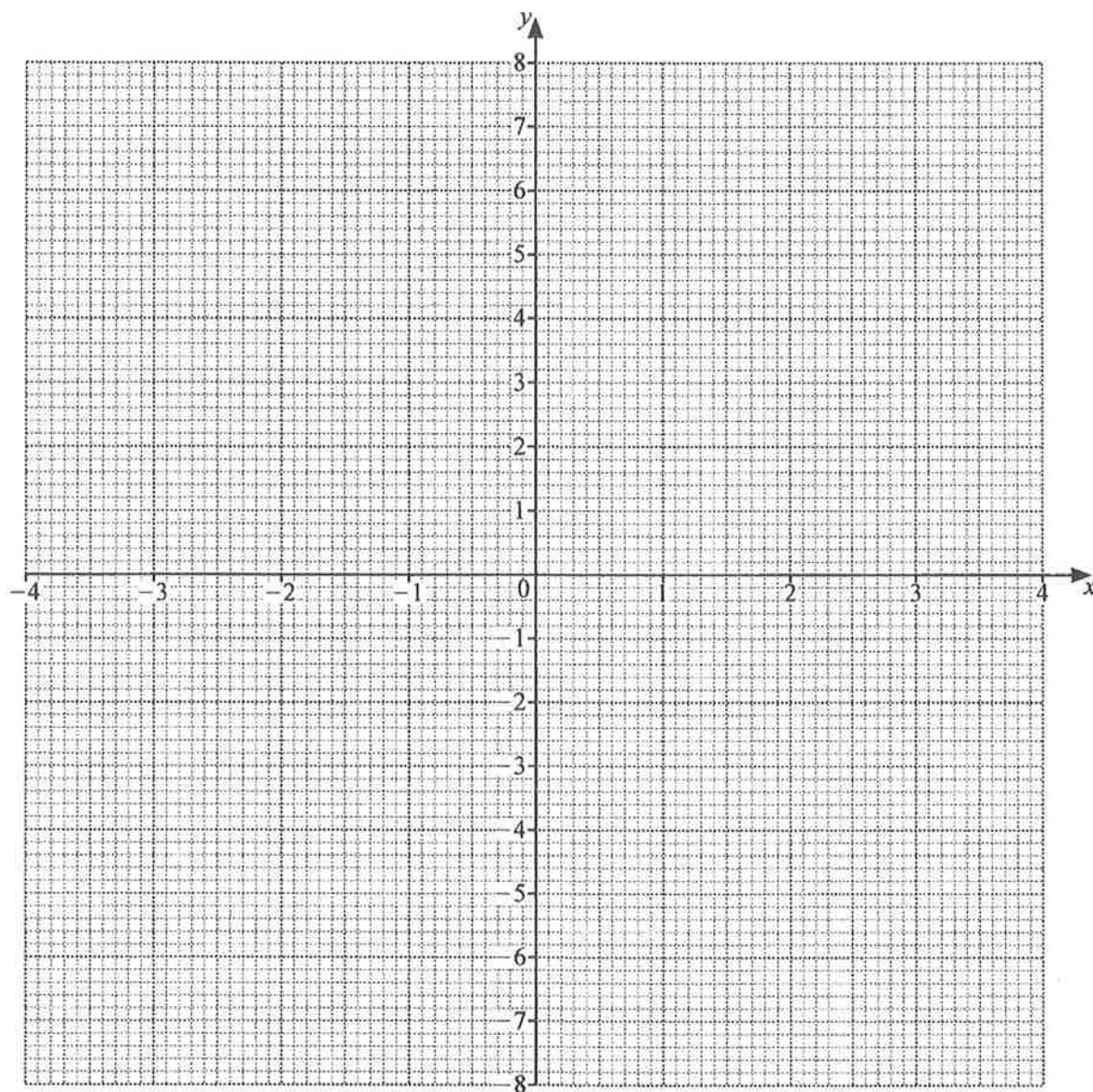
Paper 2: Intermediate

2. (a) Complete the table of values for $y = \frac{x^3}{5} - 2x + 1$.

x	-4	-3	-2	-1	0	1	2	3	4
y		1.6	3.4	2.8	1	-0.8	-1.4	0.4	5.8

[1]

- (b) On the grid, draw the graph of $y = \frac{x^3}{5} - 2x + 1$ for $-4 \leq x \leq 4$.



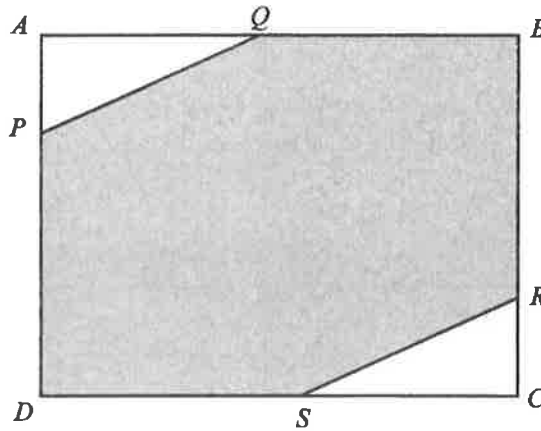
- (c) The equation $\frac{x^3}{5} - 2x + 1 = k$ has two solutions.

Use your graph to find the two possible values of k .

- (d) By drawing a suitable straight line on the grid, solve the equation $2x^3 - 25x + 20 = 0$.

[N22/II/4]

3.



$ABCD$ is a rectangle.

The area of the rectangle is 80 cm^2 .

Triangle PAQ and triangle RCS are removed from the rectangle.

$DP = QB = BR = SD = 4 \text{ cm}$.

$AD = x \text{ cm}$.

(a) Find an expression, in terms of x , for AQ .

[2]

(b) Show that the shaded area, $y \text{ cm}^2$, is given by $y = \frac{320}{x} + 4x - 16$.

[3]

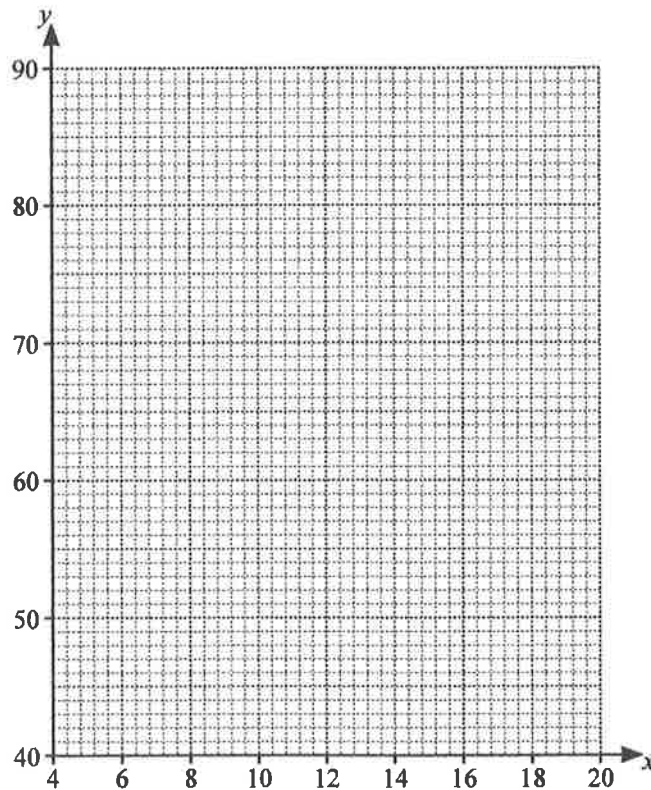
(c) Complete the table of values for $y = \frac{320}{x} + 4x - 16$.

Values are given to 1 decimal place where appropriate.

x	4	6	8	10	12	14	16	18	20
y	80	61.3	56	56	58.7	62.9	68		80

[1]

(d) On the grid, draw the graph of $y = \frac{320}{x} + 4x - 16$ for $4 \leq x \leq 20$.



[3]

(e) Use your graph to find the greatest value of x when the shaded area is equal to 60 cm^2 .

[1]

(f) Explain how the graph shows that the shaded area cannot be equal to 50 cm^2 .

[1]

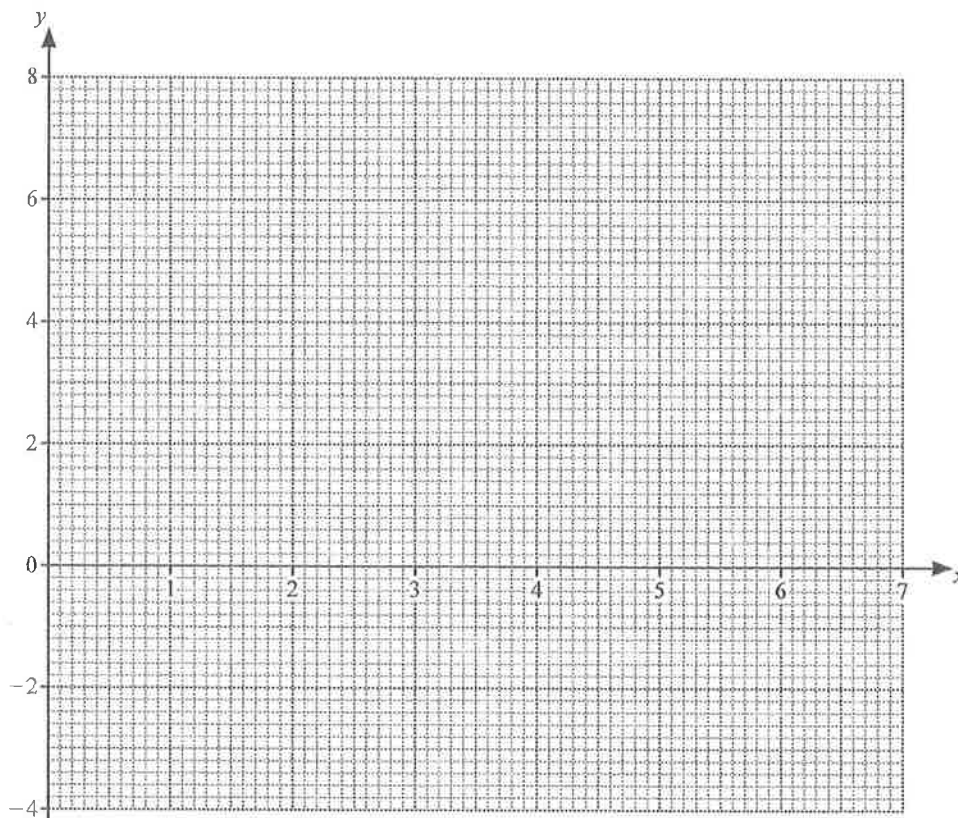
[N21/II/3]

4. (a) Complete the table of values for $y = 2x + \frac{6}{x} - 9$.
Give your answer correct to 1 decimal place.

x	0.5	1	1.5	2	3	4	5	6	7
y	4	-1	-2	-2	-1	0.5	2.2	4	

[1]

- (b) On the grid, draw the graph of $y = 2x + \frac{6}{x} - 9$ for $0.5 \leq x \leq 7$.



[3]

- (c) Use your graph to find the solutions of the equation $2x + \frac{6}{x} = 10$ in the range $0.5 \leq x \leq 7$.

[2]

- (d) (i) On the grid in part (b), draw the line $3y = 2x - 3$ for $0 \leq x \leq 7$.

[2]

- (ii) Write down the x -coordinates of the points where this line intersects the curve.

[2]

- (iii) These values of x are the solutions of the equation $2x^2 + Ax + B = 0$.

Find the value of A and the value of B .

[3]

[N20/II/3]

5. The variables x and y are connected by the equation $y = \frac{x^3}{5} - 2x + 1$.

Some corresponding values of x and y are given in the table below.

x	-4	-3	-2	-1	0	1	2	3	4
y	p	1.6	3.4	2.8	1	-0.8	-1.4	0.4	5.8

- (a) Find the value of p .

[1]

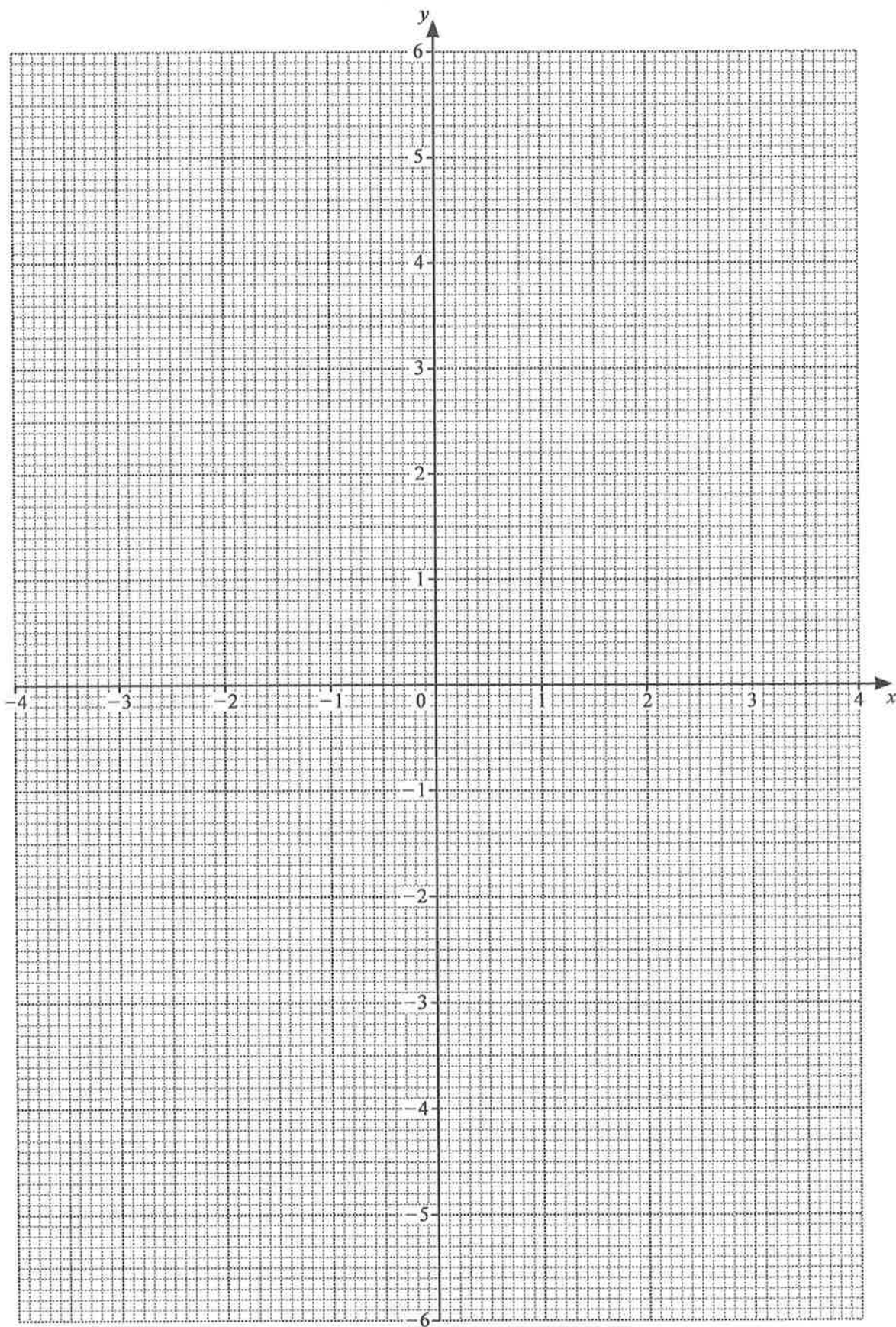
- (b) On the grid, draw the graph of $y = \frac{x^3}{5} - 2x + 1$ for $-4 \leq x \leq 4$.

[3]

- (c) Use your graph to write down an inequality in x to describe the range of values where $y > 4$.

[1]

- (d) (i) On the same grid, draw the graph of $5y + x = 10$ for $-4 \leq x \leq 4$. [2]
- (ii) Show that the points of intersection of the line and the curve give the solutions of the equation $x^3 - 9x - 5 = 0$. [2]
- (iii) Use your graphs to solve the equation $x^3 - 9x - 5 = 0$. [2]



[N19/II/4]

6. Answer the whole of this question on a sheet of graph paper.

A number of bacteria are introduced to a culture.

The number of bacteria, y , in the culture t hours after they are first introduced is given by the formula

$$y = 50 \times 2^t.$$

The table shows some corresponding values of t and y , correct to 3 significant figures.

t	0.5	1	1.5	2	2.5	3	3.5	4
y	p	100	141	200	283	400	566	800

- (a) How many bacteria are introduced to the culture at the start? [1]
- (b) Calculate the value of p . [1]
- (c) Using a scale of 4 cm to represent 1 hour, draw a horizontal t -axis for $0 \leq t \leq 4$.
Using a scale of 2 cm to represent 100 bacteria, draw a vertical y -axis for $0 \leq y \leq 800$.
On your axes, draw a graph to show the number of bacteria in the culture for $0 \leq t \leq 4$. [3]
- (d) Use your graph to find how many hours it takes for the number of bacteria to reach 500. [1]
- (e) (i) By drawing a tangent, find the gradient of the curve at (2, 200). [2]
(ii) What does this gradient represent? [1]

[N17/II/3]

7. Answer the whole of this question on a sheet of graph paper.

The variables x and y are connected by the equation $y = \frac{x^3}{2} - 5x - 2$.

Some corresponding values of x and y are given in the table below.

x	-3	-2	-1	0	1	2	3	4
y	p	4	2.5	-2	-6.5	-8	-3.5	10

- (a) Find the value of p . [1]
- (b) Using a scale of 2 cm to represent 1 unit, draw a horizontal x -axis for $-3 \leq x \leq 4$.
Using a scale of 1 cm to represent 2 units, draw a vertical y -axis for $-10 \leq y \leq 12$.
On your axes, plot the points given in the table and join them with a smooth curve. [3]
- (c) The equation $\frac{x^3}{2} - 5x = 8$ has only one solution.
Explain how this can be seen from your graph. [2]
- (d) By drawing a tangent, find the gradient of the curve at (1, -6.5). [2]
- (e) (i) On the same axes, draw the line $y = 4 - 3x$ for $-1 \leq x \leq 4$. [2]
(ii) Write down the x -coordinate of the point where this line intersects the curve. [1]
(iii) This value of x is a solution of the equation $x^3 + Ax + B = 0$.
Find the value of A and the value of B . [2]

[N16/II/5]

8. Answer the whole of this question on a sheet of graph paper.

The variables x and y are connected by the equation $y = \frac{x^2}{5} + \frac{4}{x} - 3$.

Some corresponding values of x and y , correct to two decimal places, are given in the table below.

x	0.5	1	1.5	2	2.5	3	4	5	6
y	p	1.20	0.12	-0.20	-0.15	0.13	1.20	2.80	4.87

- (a) Find the value of p . [1]
- (b) Using a scale of 2 cm to represent 1 unit on each axis, draw a horizontal x -axis for $0 \leq x \leq 6$ and a vertical y -axis for $-2 \leq y \leq 6$.
On your axes, plot the points given in the table and join them with a smooth curve. [3]
- (c) Use your graph to find the solutions to the equation $\frac{x^2}{5} + \frac{4}{x} = 3$ in the range $0 \leq x \leq 6$. [2]
- (d) By drawing a tangent, find the gradient of the curve at (4, 1.20). [2]
- (e) (i) On the same axes, draw the line with gradient -1.5 that passes through the point with coordinates (2, 1). [1]
(ii) Write down the equation of this line. [1]
(iii) Write down the coordinates of the points where the line intersects the curve. [2]

[N15/II/4]